Project Proposal: Conference Registration &Management System

DEVELOPMENT TEAM
Rod Reyes, Mona Kaur, Marina Lau, Litian Fu, Sylvia Chang

ABSTRACT
The Conference Registration and Management System is an ASP web-based application, which provides users with the ability to register for a conference and make payments online. In the backend, it will provide the conference owner with an administration panel to define, manage, and post conference details on its company website, as well as to keep track of sales generated from conferences and attendance statistics.

Our goal for this project is to develop a system that will help a corporation to effectively manage and market its conference.

Initial idea for the system architecture consists of the following five modules:

1) User Registration: allows users to create accounts and register for conferences online.
2) Payment processor: web service component for processing credit card information.
3) User Management: allows administrators to manage user information and registration.
4) Conference Management: allows an administrator to define and manage conferences.
5) Reports Central: keeps track of conference summary information.

An architecture diagram (see *figure 1*) illustrates the interaction between modules and users.

MODULE DETAILS

User registration:
Build a user registration module that allows individuals to create a user account on the website. Collect user data, such as username, name, email address, and billing information, and store it in a UserInfo table. This module will interact with a payment gateway to process credit card transactions. Subsequent transaction results will be collected from the gateway and stored in a CC_Transact table.

Payment processor:
Build a payment processor web service. Client applications will be able to use this service to process credit card billing information. The web service will store merchant account transactions in a database with pertinent billing info, including credit card details. The service will provide operations that allow merchants to retrieve specific billing or transaction data (e.g., getCCNum, getTransID, updateStreetAddress). An integration diagram (*Figure 2*) illustrates the interaction between an integration application and web service component.

User management:
Build a user management interface that allows the conference administrators to view, add, edit, delete, and sort user records. This module should also be able to request or post modified data to the payment gateway.

Conference management:
Build a conference management interface that allows administrators to view, add, edit, delete and sort event records. Each conference should have attributes such as date, time, location, price, and maximum number of registrants. Records will be stored on an EventInfo table.

Calendar management:
Build a calendar module that allows administrators to manage dates, times, and locations for conferences, conference sessions/lectures, and special events (e.g., workshops, exhibits, mixers, etc.). A possible extension is to allow conference registrants to manage their personal calendars, select which events to attend at a specific conference, or check available conference venue rooms to book their own meetings. This module will either hook into Google Calendar API, the existing .NET calendar control, or a custom calendar control.

Reports Central:
This module will be designed to track summary information, such as the number of conferences sold, sales for each conference, and attendance statistics. The business-tier layer will be defined in this module.

Figure 1 – System Architecture

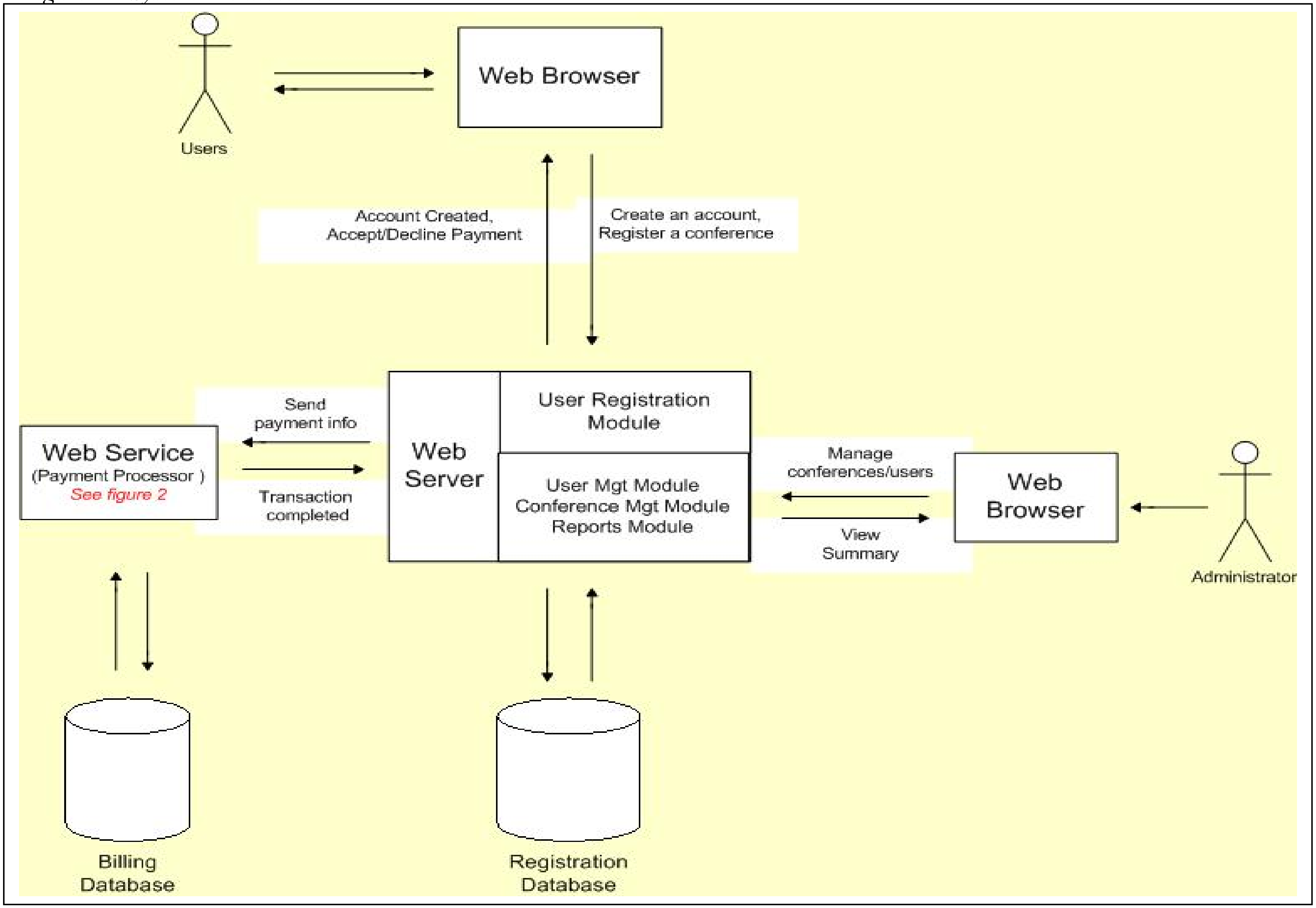
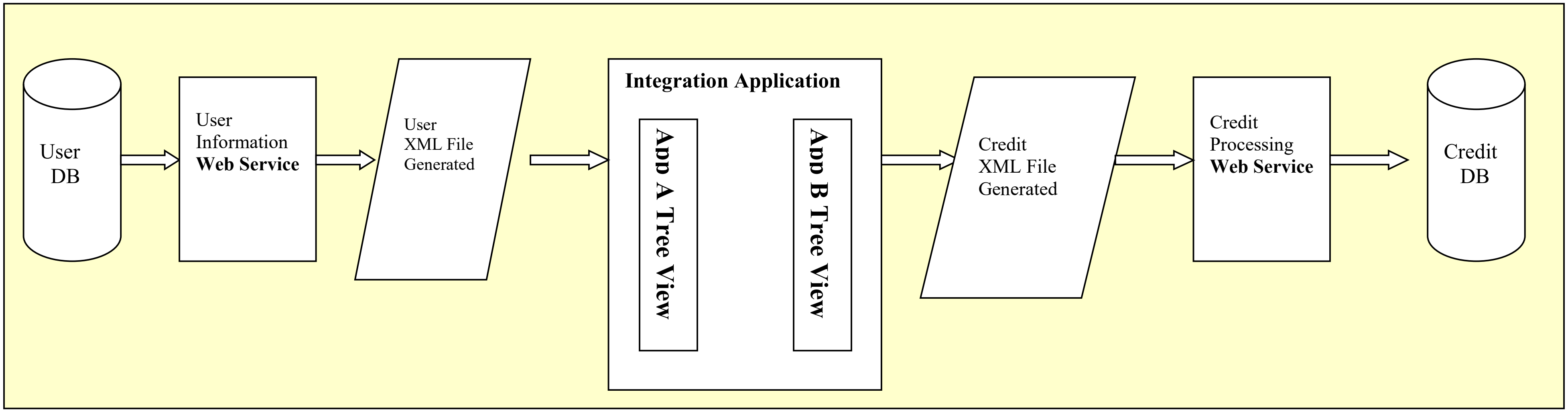


Figure 2 – Integration diagram for payment processing



Module Design: Web Service Integrator

DEVELOPMENT TEAM
Rod Reyes
Mona Kaur

ABSTRACT

Build an integrator application that will allow 2 other applications to communicate with each other. The specific applications will be: 1) payment processor web service; 2) user registration web service.

SUB-MODULE DETAILS

Application A:
Build a payment processor web service. Client applications can use this service to process credit card billing information. The web service will store merchant account transactions in a database with pertinent billing info, including credit card details. The service will provide operations that allow merchants to retrieve specific billing or transaction data (e.g., getCCNum, getTransID, updateStreetAddress).

Application B:
Build a user registration web service. This service allows client applications to manage user data. The client app will collect individual user information, such as username, name, email address, and billing information. It will provide operations that allow client apps to request or update user data (e.g., getCreditCardNumber, updateFirstName, etc.)

Integrator Application:
Build an integrator or adapter that takes data from one application and translates it for use with a second application. The integrator will compare the WSDL files from the 2 services and map the nodes accordingly to enable smooth data transfer between the apps. The integrator will use a simple tree-view interface to enable easy mapping and subsequent translation.

XML Schema Definitions

Request

Application B – UserInfo Table:	*Application A – Billing Info Table:*
User ID	First Name
FName	Last Name
LName	Billing Address 1
UserName	Billing Address 2
Password	City
Organization	State
Address1	Zip Code
Address2	Country
City	Credit Card Type
State	Credit Card Number
Country	Credit Card Name
Phone Number	Credit Card Verification
Mobile Number	Credit Card Expiration Month
Reg Date	Credit Card Expiration Year

LastChangedBy User Type	Billing Amount Merchant ID Merchant Order/Customer ID

Response

Application A – Billing Transaction Table: Billing Amount Merchant Order/Customer ID Transaction ID Transaction Status Transaction Date	*Application B – CC Transaction Table:* User ID Trans Date Trans Status Trans ID Amount